HumixAi

সম্পূর্ণ System Workflow, Features ও Implementation Status

Client-facing Functional & Technical Overview | Version 1.0

EXECUTIVE SUMMARY
HumixAi একটি AI-assisted writing, rewriting ও content-review platform। Writing Studio, document management, account, team collaboration, admin, API এবং বিভিন্ন integration-এর উল্লেখযোগ্য অংশ তৈরি হয়েছে। তবে GPTZero, SMTP, OCR, payment ও production operations-এর কিছু অংশ এখনও configuration, live testing বা expansion পর্যায়ে আছে। এই নথিতে implemented feature এবং launch-ready capability আলাদা করে দেখানো হয়েছে।

Report basis: supplied HumixAi implementation/status report dated 12 November 2026. এখানে দেওয়া operational status ও test counts উক্ত রিপোর্টের recorded evidence-এর ভিত্তিতে; এই ডকুমেন্টের জন্য নতুন live test চালানো হয়নি।

১ | Product scope ও value proposition

একটি central writing workspace থেকে user লেখা তৈরি/পেস্ট/import করতে পারে, writing signals review করতে পারে, AI-assisted rewrite নিতে পারে, original বনাম revised text তুলনা করতে পারে, document save ও export করতে পারে এবং উপযুক্ত permission থাকলে team/website/API workflow-এ ব্যবহার করতে পারে।

কার জন্য	প্রধান ব্যবহার
Individual writer / student	Draft review, rewriting, text comparison, document history ও export
Content team / editor	Shared documents, reviews, comments, brand/voice guidelines
Website operator	Consent-based selected-page monitoring ও scheduled reviews
Administrator	Accounts, quotas, jobs, provider settings, security ও audits
Developer / integration partner	Scoped HTTP API, extension client ও MCP bridge

২ | End-to-end user workflow

WORKFLOW MAP
Login / Register New or Open Document Paste / Import Text Select Language & Settings Detect AI / Humanize / Analyze & Improve Review Evidence & Diff Accept or Copy Revision Save / Share / Export

1. User account তৈরি করে অথবা login করে; প্রয়োজনে email verification ও password recovery workflow ব্যবহার করতে পারে (email delivery live setup-নির্ভর)।

2. Writing Studio-তে original text পেস্ট করে, import করে বা আগের saved document খোলে।
3. Text selection, tone, strength, length, voice reference এবং language preference নির্ধারণ করে।
4. তিনটি primary action-এর যেকোনো একটি চালায়। Results right/left panels-এ review করে।
5. Rewrite output ও diff দেখে; protected-content check পাস না করলে original অপরিবর্তিত থাকে।
6. Revision গ্রহণ/copy করে, history-তে save করে, permitted share link বা export ব্যবহার করে।

৩ | তিনটি **core button:** কীভাবে কাজ করে

Button	System workflow	User-visible output / সীমা
Detect AI — Local	Server-side writing-pattern analysis; কোনো external API key লাগে না	Pattern index, sentence evidence/highlight; এটি AI-written percentage নয় এবং trained detector নয়
Detect AI — GPTZero	Provider API request → response validation → probability/evidence/coverage display	GPTZero key configured নয়; live results যাচাই হয়নি; ন্যূনতম ৮০ words
Humanize	Full/selected text settings + voice context → model rewrite → preservation validation	Revised text, diff, copy/use revision; invalid protected changes হলে rejection
Analyze & Improve	Original scan → rewrite → protection check → revised scan → comparison	Scans অসম্পূর্ণ হলে PARTIAL; Local mode এখনও combo workflow-এ যুক্ত নয়

৪ | **Detect AI workflow:** গুরুত্বপূর্ণ ব্যাখ্যা

Local mode

Local signal মূলত বাক্যের দৈর্ঘ্যের সমতা (writing-pattern consistency) থেকে ০–১০০ index তৈরি করে। এটি AI-authorship probability, plagiarism score বা definitive human/AI verdict হিসেবে দেখানো উচিত নয়।

GPTZero mode

Provider connector, response validation, sentence evidence ও failure handling code আছে; কিন্তু API key না থাকায় runtime scan অসমর্থিত। Original ও revised—প্রতিটি input-এর আলাদা করে minimum word count পূরণ করতে হয়। Provider accuracy বা quota behaviour এখনও live verified নয়।

“PARTIAL” status-এর অর্থ

Combo-তে rewrite সফল হলেও before/after detector scan দুটি সম্পন্ন না হলে score comparison দেখানো যায় না। PARTIAL মানে text rewrite হারিয়ে গেছে, এমন নয়। Missing key, short text বা scan unavailable হলে এই অবস্থা হতে পারে।

৫ | **Humanize** ও **text safety workflow**

7. Input: full text অথবা selected passage, tone, rewrite strength, target length ও optional voice reference।
8. Processing: configured Gemini model দিয়ে rewriting; source report-এ gemini-3.1-flash-lite working configuration উল্লেখ আছে।

9. Validation: quotation, citation, numbers, key terms ও negation-sensitive content preservation check।
10. Review: original/revised diff, copy action, apply/use revision; প্রয়োজনমতো document save।
11. Failure case: protected-content violation হলে unsafe revision প্রত্যাখ্যান করা হয় এবং original রাখা হয়।

Live test evidence: একটি generic passage-এ Humanize এবং Combo rewrite/preservation কাজ করেছে; কিছু আগের generated outputs protection validation-এ rejected হয়েছে। অর্থাৎ প্রত্যেক input সফল হবে—এমন নিশ্চয়তা নেই।

৬ | Functional module inventory

Module	বর্তমানে উপলব্ধ functionality
Writing Studio	Dual editor, word count, selected rewrite, settings, revision/diff, copy, use revision
Detection UI	Auto/GPTZero/Local selector, evidence tab, sentence explanation, stale result indicator
Document management	Save, owned history/revisions, restore, deletion, expiring share links
Import & export	TXT, DOCX, supported text PDF, public URL preview; TXT/JSON/DOCX/report export, browser PDF print
Voice & quality	Session voice reference, consent-based encrypted profile; readability, grammar/meaning/guideline review
Teams	Owner/editor/reviewer roles, invitations, shared documents and comments
Accounts	Registration, login/logout, ownership, reset, signed verification, account deletion
Jobs & usage	Queue, stage status, reservations/settlement, idempotency, batch review/cleanup
Administration	/admin dashboard/backend, roles, suspension, quotas, revocation, job controls, audit trail
Localization	English, Bangla, Hindi, Urdu, Arabic, Nepali; RTL and Unicode handling
Website tools	Consented scheduled checks, sitemap preview, selected pages, pause/resume/delete, JSON reports
Writing replay	Opt-in encrypted coarse snapshots, playback/slider, event count and export
Developer tools	Scoped HTTP API, browser extension source, MCP bridge
Evaluation	Offline labelled-prediction metrics: confusion matrix, precision, recall, calibration/Brier

৭ | Admin & system operations workflow

ADMIN WORKFLOW
Admin login Review users/roles Configure quotas and provider keys Monitor queued jobs Retry / revoke / suspend as appropriate Review audit log Configure production worker and monitoring

Admin backend-এর code-এ provider credentials encrypt করে সংরক্ষণ, usage/job controls, user ownership security, token ও share-link revocation এবং audit trail অন্তর্ভুক্ত। এটি routine operational readiness-এর সমান নয়; worker supervision, backup, HTTPS এবং environment configuration আলাদাভাবে সম্পন্ন করতে হবে।

৮ | Teams, website ও integration workflow

Channel	Expected workflow	Current deployment caveat
Team collaboration	Owner invites editor/reviewer → shared document → comments/review	Roles ও comments implemented; organizational rollout আলাদাভাবে validate করা উচিত
WordPress plugin	Activate plugin → consent → import/review → API job → manual copy-back	ZIP ও client code প্রস্তুত; real WP activation এবং editor/theme compatibility untested
Website checks	Consent → sitemap/selected page preview → schedule → JSON report	Production scheduler ও real-page scans যাচাই বাকি
Browser extension / MCP	Install/connect supported client → authorize scoped workflow → API calls	Source/tests present; user environment live connect ও publication বা কি
Stripe subscription	Checkout → signed webhook sync → portal → entitlements	Price/key/webhook setup ও sandbox payment lifecycle বা কি

৯ | Technology, data handling ও operational boundaries

- Platform-এর status report-এ PHP artisan command, PHP test suite, Node client tests এবং Python evaluation tests উল্লেখ আছে। এগুলো থেকে backend/clients-এর প্রযুক্তিগত ইঙ্গিত পাওয়া যায়; পূর্ণ infrastructure diagram বা stack specification source-এ নেই।
- Consent-based encrypted personal/brand voice profiles ও opt-in writing replay আছে; writing replay complete keystroke capture বা authorship proof নয়।
- Scoped API access, user ownership, signed verification ও audit trail implemented হিসেবে reported।
- PDF support মূলত supported text-based PDFs; scanned multipage PDF OCR pipeline এখনো roadmap-এ।
- Automatic whole-site recursive crawling এবং complex/authenticated/JS-heavy import এখনও roadmap-এ।

১০ | Implementation status: কী ready, কী বাকি

Capability	Status	Required next action
Writing Studio, Humanize, docs, admin core	Implemented; selected test evidence exists	Browser E2E, accessibility, mobile/RTL QA
GPTZero	Code ready; inactive at runtime	API key + real scan, quota, response tests
SMTP verification/recovery	Code ready; mail not configured	SMTP + APP_URL + delivery validation
OCR images	Local support exists; language packs absent	Install Tesseract/language packs; real image tests
Walter similarity	Adapter ready; live status unverified	Credential/scope + live source-match

		test
Stripe	Billing flow implemented; unvalidated	Sandbox checkout/webhook/subscription lifecycle
WordPress	Plugin ZIP built; unverified in WP site	Activate and verify Classic/Block Editor compatibility
Queue/scheduler	Job code exists; temporary worker tested	Persistent supervised workers, cleanup, monitoring
Multilingual	Six catalogues; 1,039 matching keys	Native-speaker QA + RTL visual testing
Evaluator	Offline tool and synthetic tests	Independent licensed/consented data + results

১১ | Recorded test evidence (not a new certification)

Test / verification	Recorded result
PHP full suite before final prompt refinement	143 tests; 1,055 assertions passed
Focused Humanizer suite after prompt refinement	17 tests; 54 assertions passed
Combined Node suites	42 tests passed
Offline Python evaluator	4 tests passed; synthetic fixtures only
WordPress client	Isolated HTTP stub contract checks passed; not live WordPress
Live Gemini	Generic Humanize/Combo rewrite passed; some protections rejected outputs
Local HTTP	Corrected template/assets served

Not yet proven: live GPTZero accuracy, actual payment lifecycle, SMTP delivery, real OCR, production website monitoring, WordPress compatibility, complete mobile/browser UX or independently assessed multilingual quality.

১২ | Delivery roadmap ও acceptance criteria

Priority	Deliverable	Suggested acceptance / sign-off
P0 — Core consistency	Connect Local/Auto/GPTZero selection to Combo	Before/after report follows selected mode; local metric clearly labelled non-probability
P0 — Provider live verification	Configure GPTZero and run three-button scenarios	Valid 80+ word scans, short input, key missing, quota/failure paths logged
P0 — Reliability	Supervised queue/scheduler, cleanup and monitoring	Jobs survive restarts; retries, failure visibility and alerts documented
P0 — Browser quality	E2E, responsive, RTL and accessibility QA	Main actions, save/reopen, import/export, stale-state behaviour verified
P1 — Integrations	SMTP, OCR, optional Stripe/Walter, WordPress	Production-like staging tests with signed-off evidence
P1 — Release operations	HTTPS, backups, restore, logs, performance tests	Deployment checklist and recovery drill completed

P2 — Expansion	Webhooks, top-ups, Docs/LMS, stronger detector, mobile	Separate scope, cost, API dependency and success criteria agreed

১৩ | Client handover: what can be demonstrated today

12. Show Writing Studio with original/revised editor and diff using suitable generic text.
13. Demonstrate Local detector and explicitly explain that its number is a pattern index—not AI probability.
14. Demonstrate a successful Humanize rewrite and a protection-based rejected revision as two expected cases.
15. Demonstrate Combo returning PARTIAL if provider scans are unavailable; explain why.
16. Show saved history, revision reopening, supported import/export paths and applicable account/team/admin interfaces.
17. Present GPTZero, WordPress, payment, OCR, email, schedule and language evaluation as pending live activation/verification—not launched guarantees.

১৪ | Decisions required from client

- Confirm MVP launch scope: writing studio only, or team/admin/WordPress/payment in the first release?
- Provide/approve provider accounts and keys (GPTZero, SMTP, Stripe, optional Walter); credentials must be handled securely, not inserted into reports.
- Confirm hosting, domain/HTTPS, backup retention, privacy and consent requirements.
- Agree on detector accuracy claims: no unsupported “AI percentage” promise for Local mode.
- Approve acceptance tests and release gate before announcing production readiness.

১৫ | Longer-term product roadmap (not current delivery)

Proprietary calibrated multilingual detector; optional second-opinion detection; source-backed fact/citation verification; credit top-ups and advanced team analytics; outgoing webhooks/Zapier; Google Docs/LMS integrations; full WordPress rich-editor replacement; recursive crawling and complex sitemap support; scanned multipage PDF OCR; full keystroke replay; native mobile applications; and multimedia/deepfake/moderation detection. এসবকে বিদ্যমান product feature হিসেবে বিক্রয়/প্রচার করা উচিত নয়।

Source & reporting notes

Primary source: “HumixAi — কী শেষ, কী আংশিক, কী বাকি”, dated 12 November 2026, supplied by the project team. Reference paths mentioned in source: humixai-product-blueprint.md; implementation-status.md; blueprint-expansion.md; verification.md; admin-and-localization.md; detector-evaluation.md; integrations/wordpress/README.md. These referenced files were not separately supplied or independently reviewed for this document.

Prepared as a client communication and planning document. All statuses are source-reported, not an independent security audit or production certification.